

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: 1.7 MICROORGANISMS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Remember the sufuria of leftover ugali and the half-eaten chapati left uncovered on a kitchen counter in Kisumu overnight. By morning there were white, green and black fuzzy patches, a sour smell and some slimy areas. Nobody had touched the food, no insects were visible and the kitchen looked clean. Use what you learned in Lessons 1 to 8 to answer the Driving Question: Why did the ugali and chapati grow fuzzy patches overnight, and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya? Write your answers in the space under each prompt. Use full sentences, name the microorganisms and conditions you studied, and refer to the Kisumu ugali and chapati in every answer. Where a prompt asks for evidence, say what you observed or learned in class.

Part 1: What are the patches and where did they come from?

Look back at the photographs of the Kisumu ugali and chapati. (a) Are the white, green and black patches living or non-living? What type of microorganism caused them? (b) Nobody touched the food and the kitchen looked clean. Explain where the organisms came from and how they reached the food. (c) The patches were different colours. What does this tell you about the organisms that grew on the food?

(a) The patches are living. They are colonies of fungal mould. Fungi are eukaryotic organisms with cell walls made of chitin, and they grow as visible fuzzy threads when many cells grow together. (b) Microscopic fungal spores were already floating in the kitchen air. They are too small to see, so a kitchen can look clean and still contain them. This is airborne transmission: spores carried in air currents settled on the uncovered ugali and chapati. Nobody had to touch the food for the spores to land. (c) The different colours show that more than one kind (species) of mould grew on the same food, for example Rhizopus, Penicillium and Aspergillus. All of them arrived by the same airborne route, but each formed its own colour and texture of colony on the food surface.

Part 2: Why did it happen overnight?

In Lesson 7 you compared mould growth on ugali, bread and dry maize biscuit under different conditions (covered or uncovered, moist or dry, warm or cool). (a) Which three

(a) Mould growth is favoured by warmth, moisture and exposure to air. The warm, moist, uncovered ugali and bread samples showed extensive white, green and black fuzzy growth. The dry, cool, covered samples showed little or no visible growth. (b) The kitchen was warm and the ugali and chapati were moist, so the spores that landed could germinate and grow quickly. Warmth speeds up fungal enzyme activity and growth, and water lets the fungus digest the food and absorb nutrients. The food was uncovered, so spores could settle on it. All three conditions were present at the same time, so visible colonies formed by morning. (c) The dry maize biscuit showed the least growth under all conditions because it has little water. This shows that moisture is

conditions favoured mould growth, and what did the pre-incubated samples show about them? (b) Use these conditions to explain why the Kisumu ugali and chapati spoiled so quickly. (c) Why is the dry maize biscuit useful as a comparison?	needed for mould to grow, even when spores and air are present.
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Part 3: Protecting our food and our health

(a) Choose TWO ways of protecting the ugali and chapati from spoilage. For each, say which condition or transmission route it interrupts. (b) Some Kisumu households, such as Mama Adhiambo's in Nyalenda, may not have a refrigerator. Suggest low-cost options and explain why they work. (c) Name TWO traditional Kenyan food preservation methods from our lessons and explain why they work.	(a) Covering the sufuria blocks airborne spores from landing on the food, so it interrupts the airborne route. Refrigerating the food lowers the temperature below the range that suits most moulds and bacteria, so it slows germination and growth of any spores that did land. Drying also works because it removes the water the mould needs. (b) A clean cloth or a wire mesh dome over the food costs little or nothing and physically stops spores from settling. Eating the food soon after cooking, and keeping it dry, also helps. Community actions such as women's group sensitisation and school hygiene clubs can spread these low-cost ideas to households without refrigeration. (c) Drying omena and salting meat both reduce the water available, so microorganisms cannot grow well even without a refrigerator. They work because they remove one of the conditions microorganisms need.
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Part 4: The wider world of microorganisms

(a) The mould on the ugali belongs to the fungi. Name one infection caused by fungi, one by bacteria and one by viruses that you studied, and say how each is spread. (b) Why does knowing the mode of transmission help us choose the best way to prevent an infection? (c) Microorganisms are not always harmful. Give TWO beneficial roles from Lesson 6 and explain how the same group of organisms can be both a problem and useful.	(a) Fungi: ringworm (spread by direct contact) or oral thrush. Bacteria: cholera, caused by <i>Vibrio cholerae</i> and spread through contaminated water, as in communities near Lake Victoria, or tuberculosis, spread through the air by droplets. Viruses: influenza or COVID-19, spread by respiratory droplets, or HIV, spread through body fluids. (b) Each control measure works by breaking one route. Covering food stops airborne spores, hand-washing stops direct contact spread, and boiling water and cooking food thoroughly stop contaminated food and water spread. If we know the route we can pick the measure that blocks it. (c) <i>Lactobacillus</i> ferments milk to make yoghurt, and <i>Saccharomyces</i> makes mandazi and bread rise. <i>Rhizobium</i> fixes nitrogen for bean and maize farms, and <i>Penicillium</i> produces penicillin. The same group can be both harmful and useful depending on where it grows. Fungi spoil food left uncovered, yet yeast and <i>Penicillium</i> are valuable when used in controlled conditions.
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Part 5: Answer the Driving Question

Write a short message of about 5 to 6 sentences for households in your own locality. In it, explain why the ugali and chapati grew fuzzy patches overnight and give TWO pieces of advice to protect food and health. Include at least one piece of evidence from our lessons.

Dear neighbours, if you leave ugali or chapati uncovered overnight, it can grow white, green and black fuzzy patches. These patches are living mould, a type of fungus. Tiny mould spores are always floating in the air, even in a clean kitchen, and they settle on the food. Warm, moist food left uncovered is the perfect place for them to grow, and in our class investigation the warm, moist, uncovered samples grew the most mould. Please cover your food with a clean cloth or wire mesh dome, and keep it cool or dry where you can. You can also eat food soon after it is cooked and wash your hands before eating. These simple steps protect our food, save money and protect our families' health.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explanation of the phenomenon (what and where from)	Clearly identifies the patches as living fungal mould colonies, explains airborne spores as the source and route, and explains the colour variation as different species.	Identifies the patches as fungal mould and says spores came from the air, but gives limited explanation of the colour variation.	Says the patches are germs or dirt, or does not explain how the organisms reached the food.
Use of evidence on growth conditions	Names warmth, moisture and air exposure, uses the Lesson 7 observations, and links all three to the overnight spoilage in Kisumu.	Names the conditions and links most of them to the phenomenon, with limited use of investigation evidence.	Names one condition or gives a vague link to the phenomenon, with no evidence.
Prevention and control reasoning	Matches each control measure to the route or condition it interrupts and suggests accessible, low-cost options for Kenyan households.	Gives suitable control measures with some explanation of how they work.	Lists measures without explaining why they work.
Scientific knowledge of microorganisms	Accurately describes infections, transmission routes and beneficial and harmful roles, and shows how the same group can be both.	Mostly accurate examples with minor gaps in transmission or the dual role.	Few or inaccurate examples.
Communication and Kenyan context	Clear, well-organised message for the community that uses accurate scientific terms and local examples.	Clear message with some scientific terms and some local context.	Message is unclear or lacks scientific terms and local context.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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